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Getting Help

Having established a solid foundation in Linux's structure, its various distributions, and the purpose of the shell, we're now prepared to put this knowledge into action. It's time to dive in, using commands directly in the terminal, as well as learning how to seek help when we encounter unfamiliar ones.

We will always stumble across tools whose optional parameters we do not know from memory or tools we have never seen before. Therefore it is vital to know how we can help ourselves to get familiar with those tools. The first two ways are the man pages and the help functions. It is always a good idea to familiarize ourselves with the tool we want to try first. We will also learn some possible tricks with some of the tools that we thought were not possible. In the man pages, we will find the detailed manuals with detailed explanations.

First Command:

```
athane@htb[/htb]$ ls
```

```
shellsession
```

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The `ls` command in Linux and Unix systems is used to list the files and directories within the current folder or any specified directory, allowing you to see what's inside and manage files more effectively. Like most Linux commands, `ls` comes with additional options and features that help you filter or format the output to display exactly what you want. To discover which options a tool or command offers, there are several ways to get help. One such method is using the `man` command, which displays the manual pages for commands and provides detailed information about their usage.

Syntax:

```
athane@htb[/htb]$ man <tool>
```

shellsession

Let us have a look at an example and get help for the `ls` command:

Example:

```
athane@htb[/htb]$ man ls
```

shellsession

LS(1)

User Commands

LS(1)

shellsession

NAME

`ls - list directory contents`

SYNOPSIS

`ls [OPTION]... [FILE]...`

DESCRIPTION

List information about the FILES (the current directory by default). Sort entries alphabetically if none of `-cftuvSUX` nor `--sort` is specified.

Mandatory arguments to long options are mandatory for short options too.

`-a, --all`

do not ignore entries starting with `.`

`-A, --almost-all`

do not list implied `.` and `..`

`--author`

Manual page `ls(1)` line 1 (press `h` for help or `q` to quit)

After looking at some examples, we can also quickly look at the optional parameters without browsing through the complete documentation. We have several ways to do that.

Syntax:

```
athane@htb[/htb]$ <tool> --help
```

Example:

shellsession

```
athane@htb[/htb]$ ls --help
```

Usage: ls [OPTION]... [FILE]...

List information about the FILES (the current directory by default).

Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.

-a, --all	do not ignore entries starting with .
-A, --almost-all	do not list implied . and ..
--author	with -l, print the author of each file
-b, --escape	print C-style escapes for nongraphic characters
--block-size=SIZE	with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below
-B, --ignore-backups	do not list implied entries ending with ~
-c	with -lt: sort by, and show, ctime (time of last modification of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first
-C	list entries by columns

<SNIP>

Some tools or commands like `curl` provide a short version of help by using `-h` instead of `--help`:

Syntax:

```
athane@htb[/htb]$ <tool> -h
```

shellsession

Example:

```
athane@htb[/htb]$ curl -h
```

shellsession

Usage: curl [options...] <url>

--abstract-unix-socket <path> Connect via abstract Unix domain socket
--anyauth Pick any authentication method
-a, --append Append to target file when uploading
--basic Use HTTP Basic Authentication
--cacert <file> CA certificate to verify peer against
--capath <dir> CA directory to verify peer against
-E, --cert <certificate[:password]> Client certificate file and password
<SNIP>

As we can see, the results from each other do not differ in this example. Another tool that can be useful in the beginning is `apropos`. Each manual page has a short description available within it. This tool searches the descriptions for instances of a given keyword.

Syntax:

```
athane@htb[/htb]$ apropos <keyword>
```

Example:

**Linux Fundamentals**

0%



```
sudo (8) - execute a command as another user
sudo.conf (5) - configuration for sudo front end
sudo_plugin (8) - Sudo Plugin API
sudo_root (8) - How to run administrative commands
sudoedit (8) - execute a command as another user
sudoers (5) - default sudo security policy plugin
sudoreplay (8) - replay sudo session logs
visudo (8) - edit the sudoers file
```

Another useful resource to get help if we have issues to understand a long command is:

<https://explainshell.com/>

Next, we'll be covering a large number of commands, many of which may be new to you. However, you now know how to seek help with any command you're unfamiliar with, or unsure about its options. Also, we highly encourage you to explore your curiosity, taking as much time as needed to tinker and experiment with the tools presented. It will always be time well spent.



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